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Slit nozzle and method for manufacturing high-silicon steel strip

Abstract

A slit nozzle having a double-tube structure and a method for manufacturing a high-silicon steel strip having a small variation in Si concentration depending on the position in the width direction of the steel strip. The slit nozzle has a double-tube structure, in which a flow-control plate which closes a gap between an inner tube and an outer tube is disposed between an open end of the inner tube and an end of a delivery port, and in which an opening is formed in a plane in which the flow-control plate is disposed only in a range of the flow-control plate of 27.5° or more and 332.5° or less in terms of a central angle with respect to a reference line L1 passing through the axis of the outer tube and the central position in the width direction of the delivery port.

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Background/Summary

TECHNICAL FIELD

(1) This application relates to a slit nozzle and a method for manufacturing a high-silicon steel strip. In more detail, the application relates to a slit nozzle having a double-tube structure with which it is possible to decrease a variation in the flow rate of a blown gas depending on the position in the axis direction and to a method for manufacturing a high-silicon steel strip which utilizes the slit nozzle.

BACKGROUND

(2) Examples of a known method for industrially manufacturing a high-silicon steel strip having a Si content of 4 mass % or more include a siliconizing treating method. This manufacturing method is a method in which a high-silicon steel strip is continuously manufactured by blowing a treatment gas containing silicon tetrachloride (SiCl_4) onto a thin steel strip having a Si concentration of less than 4 mass % at a high temperature to make Si permeate into the steel strip and by performing a heat treatment on the steel strip so that the Si which has permeated into the surface of the steel strip is diffused in the thickness direction.

(3) Examples of a known method for blowing the treatment gas include a method in which slit nozzles having a delivery port (slit) for the treatment gas are arranged on both the front-surface side and back-surface side of the steel strip in a siliconizing treating furnace and the treatment gas is blown through the gas delivery ports onto the steel strip (for example, refer to Patent Literature 1).

(4) In addition, examples of a known slit nozzle **10** include a slit nozzle which is illustrated in a sectional view of FIG. **10** and which has a double-tube structure of an outer tube **20** having a delivery port (slit) **21** for the treatment gas and an inner tube **30** having one end through which the treatment gas is fed and another end which is open in the outer tube **20** (for example, refer to Patent Literature 2).

(5) As illustrated in FIG. **10**, in the case where the treatment gas is blown onto the steel strip by using such a slit nozzle **10**, the flow rate of the treatment gas increases with an increase in the distance from the gas feeding port **31**. Therefore, in the case of this method, there is a variation in the Si concentration depending on the position in the width direction of the steel strip due to a variation in the flow rate of the treatment gas blown through the slit nozzle. As a result, there is a problem in that a shape defect and a variation in magnetic properties occur in the steel strip due to a difference in lattice constant.

(6) To solve such a problem, there is a known method for manufacturing a high-silicon steel strip in

which, as illustrated in FIG. 11, by feeding the treatment gas to slit nozzles **10** adjacent to each other in the furnace length direction in such a manner that the gas is fed in opposite directions alternately, it is possible to decrease a variation in the Si concentration depending on the position in the width direction of a steel strip **11** (for example, refer to Patent Literature 2 and Patent Literature 3).

(7) However, even in the case of the manufacturing methods according to Patent Literature 2 and Patent Literature 3 described above, it is not possible to sufficiently decrease a variation in the Si concentration depending on the position in the width direction of the high-silicon steel strip. Therefore, there is a demand for a slit nozzle having a double-tube structure which decreases a variation in the flow rate of a blown gas depending on the position in the axis direction as compared to the conventional slit nozzles.

CITATION LIST

Patent Literature

(8) PTL 1: Japanese Unexamined Patent Application Publication No. 62-227078 PTL 2: Japanese Unexamined Patent Application Publication No. 8-176793 PTL 3: Japanese Unexamined Patent Application Publication No. 5-9704

SUMMARY

Technical Problem

(9) The disclosed embodiments have been completed in view of the situation described above, and an object is to provide a slit nozzle having a double-tube structure with which it is possible to decrease a variation in the flow rate of a blown gas depending on the position in the axis direction and to provide a method for stably manufacturing a high-silicon steel strip having a small variation in the Si concentration depending on the position in the width direction of the steel strip.

Solution to Problem

(10) In the process of the investigations regarding the reasons of the problem described above and the like which were conducted to solve the problems described above, the present inventors found that, by arranging a predetermined flow-control plate between an open end of an inner tube of a double-tube structure and the end of a delivery port for the treatment gas, it is possible to decrease a variation in the flow rate of a blown gas depending on the position in the axis direction, resulting in completion of the disclosed embodiments.

(11) The subject matter of the disclosed embodiments for solving the problems described above is as follows.

(12) [1] A slit nozzle having a double-tube structure including an outer tube having a delivery port for a treatment gas in an axis direction and a closed end, and an inner tube having a feeding port for the treatment gas on one end and an open end that is another end inside the closed end of the outer tube, the treatment gas being fed through the feeding port and blown through the delivery port, the slit nozzle including a flow-control plate which is disposed between the open end of the inner tube and an end, near the open end, of the delivery port and which closes a gap between the inner tube and the outer tube, wherein an opening is formed in a plane in which the flow-control plate is disposed only in a range of the flow-control plate of 27.5° or more and 332.5° or less in terms of a central angle with respect to a reference line passing through an axis of the outer tube and a central position in a width direction of the delivery port.

(13) [2] The slit nozzle according to item [1], in which the flow-control plate is disposed at the open end of the inner tube.

(14) [3] The slit nozzle according to item [1] or [2], in which the opening is formed symmetrically with respect to the reference line in the plane in which the flow-control plate is disposed.

(15) [4] A method for manufacturing a high-silicon steel strip using a siliconizing treating method utilizing the slit nozzle according to any one of items [1] to [3], the method including: arranging a plurality of the slit nozzles in a threading direction of a steel strip in a siliconizing treating furnace such that slit nozzles or groups of slit nozzles adjacent to each other in the threading direction are

arranged so that feeding ports for the treatment gas of the slit nozzles face opposite directions from each other, and feeding the treatment gas containing silicon tetrachloride (SiCl_4) through the feeding ports for the treatment gas into the slit nozzles and blowing the treatment gas through delivery ports for the treatment gas of the slit nozzles onto the steel strip transported.

Advantageous Effects

(16) According to the disclosed embodiments, it is possible to decrease a variation in the flow rate of a blown gas depending on the position in the axis direction in a slit nozzle having a double-tube structure. In addition, by using such a slit nozzle, it is possible to stably manufacture a high-silicon steel strip having a small variation in the Si concentration depending on the position in the width direction of the steel strip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view illustrating slit nozzles and a steel strip in a siliconizing treating furnace.
- (2) FIG. 2 is a sectional view illustrating an example of a slit nozzle according to an embodiment.
- (3) FIG. 3 is a sectional view along line III-III in FIG. 2 illustrating an example of a flow-control plate.
- (4) FIG. 4 is a sectional view illustrating a range of the flow-control plate of 27.5° or more and 332.5° or less in terms of a central angle with respect to the reference line.
- (5) FIG. 5 is a schematic diagram illustrating an example of a continuous line for manufacturing a high-silicon steel strip.
- (6) FIG. 6 is a schematic diagram illustrating a case where adjacent slit nozzles are arranged so that their feeding ports for the treatment gas face opposite directions from each other.
- (7) FIG. 7 is a graph illustrating the evaluation results of Example 1.
- (8) FIG. 8 is a graph illustrating the evaluation results of Example 2.
- (9) FIG. 9 is a graph illustrating the evaluation results of Example 3.
- (10) FIG. 10 is a sectional view illustrating a slit nozzle of the related art.
- (11) FIG. 11 is a schematic diagram according to the related art illustrating a case where adjacent slit nozzles are arranged so that their feeding ports for the treatment gas face opposite directions from each other.

DETAILED DESCRIPTION

- (12) Hereafter, embodiments will be described with reference to the drawings. However, it is not intended that the disclosed embodiments be limited to the illustrated examples. In addition, the flow directions of the treatment gas which flows in the slit nozzle or which is fed into the slit nozzle are indicated by the arrows in the drawings.
- (13) According to embodiments, a slit nozzle which has a double-tube structure including an outer tube and an inner tube and in which a treatment gas is blown through a delivery port. Here, although examples in which a siliconizing treatment is performed, that is, Si is made to permeate into a steel strip by using the slit nozzle, are described in the description of the embodiment below, the disclosed embodiments are not limited thereto and may be used for other purposes as long as the effects of the disclosed embodiments is realized. For example, the slit nozzle may be used when a ceramic film such as a TiN film is formed on a steel sheet or when various chemical vapor deposition treatments are performed not only on a steel sheet but also on an aluminum sheet, a copper sheet, or the like.
- (14) In the case of a siliconizing treatment utilizing a slit nozzle 10, a slit nozzle 10 having a delivery port (slit) 21 for a treatment gas is arranged on each of the front-surface side and back-surface side of a steel strip 11 in a siliconizing treating furnace and a treatment gas containing

silicon tetrachloride (SiCl₄) is blown through the delivery port **21** of the slit nozzle **10** onto the steel strip **11** at a high temperature to make Si permeate into the steel strip (refer to FIG. **1**). In addition, a heat treatment is performed on the steel strip so that the Si which has permeated into the surface of the steel strip is diffused in the thickness direction, thereby continuously manufacturing a high-silicon steel strip.

(15) FIG. **2** is a sectional view of the slit nozzle **10** illustrated in FIG. **1**. The slit nozzle **10** illustrated in FIG. **2**, that is, an example of the slit nozzle according to the disclosed embodiments which is shown in a sectional view, has a double-tube structure including an outer tube **20** and an inner tube **30**. In addition, a flow-control plate **40**, which controls the flow of the treatment gas in the slit nozzle **10**, is disposed inside the slit nozzle **10**.

(16) The outer tube **20** has a delivery port (slit) **21** for a treatment gas in the axis direction D2. In addition, one end (end on the left-hand side of FIG. **2**) of the outer tube **20** is closed. In addition, in the case of the example illustrated in FIG. **2**, the outer tube **20** has a hole, which has a diameter corresponding to the outer diameter of the inner tube **30** so that it is possible to dispose the inner tube **30** inside the outer tube, at the other end (end on the right-hand side of FIG. **2**) of the outer tube **20** with other part of the other end being closed. However, the other end (end on the right-hand side of FIG. **2**) of the outer tube **20** does not necessarily have to be closed.

(17) The inner tube **30** is disposed inside the outer tube **20**. In addition, the inner tube **30** has a feeding port **31** for a treatment gas at one end (on the right-hand side of FIG. **2**) and another end (on the left-hand side of FIG. **2**) of the inner tube **30** is open inside the one end, which is closed, of the outer tube **20**.

(18) In addition, as illustrated in FIG. **2**, when a treatment gas is blown into the slit nozzle **10** through the feeding port **31**, the treatment gas is transported through the inside of the inner tube **30** and blown through an open end **32** of the inner tube **30** into the inside of the one end, which is closed, of the outer tube **20**. Subsequently, the treatment gas is made to turn back inside the outer tube **20**, transported through a gap between the outer tube **20** and the inner tube **30**, and, finally, blown out through the delivery port **21** of the outer tube **20**.

(19) In the case of the example illustrated in FIG. **2**, the flow-control plate **40** is disposed at the open end of the inner tube **30** so as to close the gap between the inner tube **30** and the outer tube **20**. Here, the position at which the flow-control plate **40** is disposed is not limited to this example, and the flow-control plate **40** may be disposed at any position within the range indicated by B1 between the open end **32** and the end of the delivery port **21** near the open end **32** (refer to FIG. **2**). However, it is preferable that the flow-control plate **40** be disposed at the open end of the inner tube **30** described above from the viewpoint of manufacturing efficiency.

(20) In addition, FIG. **3** is a sectional view along line III-III in FIG. **2** illustrating the plane in which the flow-control plate **40** is disposed with the delivery port **21** being projected onto the plane in which the flow-control plate **40** is disposed. As illustrated in FIG. **3**, in the flow-control plate **40**, openings **41** are formed in the plane in which the flow-control plate **40** is disposed only in a range of the flow-control plate **40** of 27.5° or more and 332.5° or less in terms of a central angle with respect to a line (hereinafter, simply referred to as “reference line L.sub.0”) passing through the axis C1 of the outer tube **20** and the central position in the width direction of the delivery port **21**.

(21) In addition, it is preferable that a width W2, which is the width in a direction perpendicular to the axis direction D2 of the outer tube **20** (refer to FIG. **3**), be 5 mm or more and 20 mm or less from the viewpoint of effectively realizing the effects of the disclosed embodiments. In addition, it is preferable that the width W2 be 15% or less of the outer diameter of the outer tube **20** from the viewpoint of effectively realizing the effects of the disclosed embodiments.

(22) Here, the range of the flow-control plate **40** of 27.5° or more and 332.5° or less in terms of the central angle with respect to the reference line L.sub.0 is surrounded by the chain line L2 in FIG. **4**.

(23) In addition, in FIG. **3** and FIG. **4**, a line L.sub.27.5 indicating the position at an angle of 27.5° in terms of the central angle with respect to the reference line L.sub.0 and a line L.sub.332.5

indicating the position at an angle of 332.5° in terms of the central angle with respect to the reference line L.sub.0 are represented by dotted lines.

(24) Hereafter, the flow of the treatment gas in the slit nozzle **10**, which is controlled by the flow-control plate **40**, will be described.

(25) As described above, the openings **41** are formed only in the range of the flow-control plate **40** of 27.5° or more and 332.5° or less in terms of the central angle with respect to the reference line L.sub.0. That is, the range of 0° or more and less than 27.5° in terms of the central angle with respect to the reference line L.sub.0 and the range of larger than 332.5° and less than 360° in terms of the central angle with respect to the reference line L.sub.0 are completely closed by the flow-control plate **40**. As a result, when the treatment gas which has been blown through the open end **32** of the inner tube **30** into the outer tube **20** is made to turn back inside the outer tube **20** and transported toward the delivery port **21**, there is a decrease in the flow rate of the treatment gas, which is supposed to pass through the position at which the flow-control plate **40** is disposed, due to collision with the flow-control plate **40**. Therefore, it is possible to decrease the flow rate of the treatment gas (indicated by the arrow G1 in FIG. **10**) which is transported to the delivery port **21** for the treatment gas via the lower part of the inner tube **30** and which causes a variation in the amount of the blown treatment gas in the slit nozzle **10** according to the conventional art. As a result, it is considered to be possible to decrease a variation in the flow rate of the gas blown through the delivery port **21** depending on the position in the axis direction D2.

(26) In addition, in the case of the example illustrated in FIG. **3**, three openings **41** are formed in the flow-control plate **40**. Here, the number of the openings **41** may be changed, and, for example, one, two, or four or more openings **41** may be formed.

(27) It is preferable that the openings **41** be formed symmetrically with respect to the reference line L.sub.0 in the plane in which the flow-control plate **40** is disposed. As a result, it is possible to decrease a variation in the flow rate of the blown gas depending on the position in the axis direction D2.

(28) In addition, when the area fraction R of the openings **41** is defined by the equation below, it is preferable that the area fraction R be 55% or more and 75% or less from the viewpoint of achieving sufficient strength of the flow-control plate **40** while effectively realizing the effects of the disclosed embodiments.

(29) area fraction $R = \text{area of openings} / \text{area of range of the flow-control plate } 40 \text{ of } 27.5^\circ \text{ or more and } 332.5^\circ \text{ or less in terms of the central angle with respect to the reference line L.sub.0 (the area of the region surrounded by the chain line L2 in FIG. 4)}$

(30) Hereafter, the method for manufacturing the high-silicon steel strip (steel strip having a Si content of 4 mass % or more) according to the disclosed embodiments will be described. In the method for manufacturing the high-silicon steel strip according to the disclosed embodiments, a siliconizing treating method utilizing the slit nozzle **10** according to the disclosed embodiments is used.

(31) FIG. **5** illustrates an example of a continuous line for manufacturing a high-silicon steel strip using a siliconizing treating method. In this manufacturing line, a steel strip **11** (for example, Si steel strip having a Si concentration of 3 mass %), which has been fed from a payoff reel **101**, is transported through a cleaning apparatus **102**, heated thereafter to a siliconizing temperature or to a temperature near the siliconizing temperature in a heating zone **103** in non-oxidizing atmosphere, and then transported into a siliconizing treating furnace **104**.

(32) In the siliconizing treating furnace **104**, plural slit nozzles **10** according to the disclosed embodiments are arranged at intervals in the longitudinal direction of the furnace (threading direction D1). In the siliconizing treating furnace **104**, a treatment gas containing a reactant gas, that is, silicon tetrachloride (SiCl_4), is blown through the slit nozzles **10** onto both surfaces of the steel strip **11**. As a result of the blown SiCl_4 reacting with Fe contained in the steel strip **11**, there is an increase in the amount of Si in the surface layer of the steel strip **11**.

(33) Subsequently, the steel strip **11** is transported into a diffusion soaking zone **105**, and subjected to a diffusion heat treatment, in which Si is diffused in the thickness direction under a non-oxidizing atmosphere which does not contain SiCl₄. After having been cooled in a cooling zone **106**, the steel strip **11** is coated with an insulating film by using an insulating film coater **107** and an oven **108** and then coiled to a tension reel **109** as a product steel strip (for example, high-silicon steel strip having a Si content of 6.5 mass %).

(34) In addition, in the case of the method for manufacturing the high-silicon steel strip according to the disclosed embodiments, it is preferable that, in the siliconizing treating furnace **104**, the plural slit nozzles **10** be arranged in the threading direction **D1** of the steel strip **11** in the siliconizing treating furnace **104** such that slit nozzles **10** or groups of slit nozzles adjacent to each other in the threading direction **D1** are arranged so that their feeding ports **31** for the treatment gas face opposite directions from each other (refer to FIG. 6). Here, the expression a “group of slit nozzles” denotes a group including two or more slit nozzles **10**. Here, although FIG. 6 illustrates only one surface side of the steel strip **11**, the slit nozzles **10** are similarly arranged on the other surface side.

(35) The method for manufacturing a high-silicon steel strip has a process of feeding a treatment gas containing silicon tetrachloride (SiCl₄) through the feeding ports **31** for the treatment gas into the slit nozzles **10** arranged as described above and blowing the treatment gas through the delivery ports **21** of the slit nozzles **10** onto the transported steel strip **11**.

(36) Here, in the case of the slit nozzles **10** according to the disclosed embodiments, since a variation in the flow rate of the treatment gas blown through the delivery ports **21** depending on the position in the axis direction **D2** is small, the slit nozzles **10** may be arranged so that the feeding ports **31** for the treatment gas face the same direction. However, when the slit nozzles **10** are arranged so that the feeding ports **31** face opposite directions from each other as described above, it is possible to further decrease a variation in the flow rate of the blown gas depending on the position in the axis direction **D2** (a direction perpendicular to the threading direction **D1**) as the total effect of the plural slit nozzles **10** arranged in the siliconizing treating furnace. Therefore, by using the siliconizing treating method, it is possible to stably manufacture a high-silicon steel strip having a small variation in the Si concentration depending on the position in the width direction of the steel strip.

(37) It will be appreciated that the above-disclosed features and functions, or alternatives thereof, may be desirably combined into different systems or methods. Also, various alternatives, modifications, variations or improvements may be subsequently made by those skilled in the art, and are also intended to be encompassed by the disclosed embodiments. As such, various changes may be made without departing from the spirit and scope of this disclosure.

EXAMPLES

(38) Although the disclosed embodiments will be specifically described in accordance with examples hereafter, the disclosed embodiments are not intended to be limited to the examples.

Example 1: Evaluating the Positions of Openings in a Flow-Control Plate

(39) <Manufacturing Slit Nozzle **1**>

(40) An outer tube (having an inner diameter of 120 mm, an outer diameter of 140 mm, and a delivery port for a treatment gas having a size of 70 mm (in the axis direction)×10 mm (in a direction perpendicular to the axis direction)), an inner tube (having an inner diameter of 60 mm and an outer diameter of 70 mm), and a flow-control plate **1** as described below were prepared. Subsequently, the inner tube is arranged inside the outer tube so that the outer tube and the inner tube had an identical axis, the flow-control plate **1** is disposed at the open end of the inner tube, and a slit nozzle **1** as illustrated in FIG. 2 was thus manufactured. Here, the distance between the closed end of the outer tube and the open end of the inner tube was set to be 25 mm.

(41) Three openings as described below were formed in the flow-control plate **1**. In addition, all the three openings had an identical size. Each of the three openings in the flow-control plate **1** had a

width W1 of 10 mm and a central angle A1 of 50° (refer to FIG. 4). In addition, the openings were located respectively in the ranges of 35° or more and 85° or less, 155° or more and 205° or less, and 275° or more and 325° or less in terms of the central angle with respect to the reference line L.sub.0 in the plane in which the flow-control plate is disposed.

(42) <Manufacturing Slit Nozzles 2 Through 6>

(43) Flow-control plates 2 through 6, whose openings had the central angles with respect to the reference line L.sub.0 which were different from those of the flow-control plate 1 and which are given in Table 1, were manufactured. In addition, slit nozzles 2 through 6 were manufactured by using the same manufacturing method as that for the slit nozzle 1, except that the flow-control plates 2 through 6 were used instead of the flow-control plate 1.

(44) Here, the width W1 and the central angle A1 of the openings of the flow-control plates 2 through 6 were the same as those of the flow-control plate 1. Therefore, the flow-control plates 1 through 6 were different from each other only in terms of the positions of the openings and had the same total area of the three openings.

(45) TABLE-US-00001

TABLE 1	Flow-control Opening Width	Opening Angle	Central Angle with Respect to Plate No.	W1 (mm)	A1 (°)	Reference Line L.sub.0	Note
1	10	50	35° or more and 85° or less	Example	155° or more and 205° or less	275° or more and 325° or less	2
2	10	50	30° or more and 80° or less	Example	155° or more and 205° or less	280° or more and 330° or less	3
3	10	50	27.5° or more and 77.5° or less	Example	155° or more and 205° or less	282.5° or more and 332.5° or less	4
4	10	50	25° or more and 75° or less	Comparative	155° or more and 205° or less	Example	285° or more and 335° or less
5	10	50	15° or more and 65° or less	Comparative	155° or more and 205° or less	Example	295° or more and 345° or less
6	10	50	7.5° or more and 57.5° or less	Comparative	155° or more and 205° or less	Example	302.5° or more and 352.5° or less

(46) In addition, when an area fraction R is defined by the equation below, each of the area fractions R of the flow-control plates 1 through 3, which are the examples of the disclosed embodiments, was 62.3%.
$$R = \frac{\text{area of openings}}{\text{area of range of the flow-control plate}}$$
 of 27.5° or more and 332.5° or less in terms of the central angle with respect to the reference line L.sub.0 (the area of the region surrounded by the chain line L2 in FIG. 4)

<Determining and Evaluating the Flow Rate of a Blown Treatment Gas>

(47) While a treatment gas was fed into the slit nozzles 1 through 6 through their feeding ports at a flow rate of 2.3 m/sec, the flow rate of a treatment gas blown from their delivery ports for the treatment gas was determined. When the flow rate of the treatment gas was determined, nitrogen was used as the treatment gas.

(48) FIG. 7 is a graph illustrating the evaluation results, that is, the relationship between the position in the axis direction (cm) and the flow rate of the treatment gas (m/sec). Here, regarding the position in the axis direction (cm) measured along the horizontal axis, the point of 0 cm corresponds to the end of the delivery port on the side of the feeding port for the treatment gas. In addition, the point of 70 cm corresponds to the end of the delivery port on the side of the open end of the inner tube.

(49) From the results illustrated in the graph in FIG. 7, it was clarified that, in the slit nozzles 1 through 3 where the openings are formed only in the range of the flow-control plate 40 of 27.5° or more and 332.5° or less in terms of the central angle with respect to the reference line L.sub.0, it was possible to maintain the flow rate of the treatment gas within a range of 2.28 m/sec to 2.32 m/sec in the range of 0 cm to 70 cm in terms of the position in the axis direction. Therefore, it was clarified that, in the slit nozzles 1 through 3, it is possible to decrease a variation in the flow rate of the gas blown through the delivery port depending on the position in the axis direction D2.

Example 2: Evaluating the Flow Rate of the Treatment Gas when Slit Nozzles are Arranged so that Feeding Ports Face Opposite Directions from Each Other

(50) Two of slit nozzles 3 (example of the disclosed embodiments) used in Example 1 were arranged for a steel strip so that their feeding ports for the treatment gas face different (opposite)

directions from each other. Here, two of the slit nozzles **1** were arranged so that positions of their delivery ports (slits) are the same in the threading direction of the steel strip.

(51) In addition, while the treatment gas was fed through the feeding port for the treatment gas into one of the two arranged slit nozzles at a flow rate of 1.5 m/sec and into the other at a flow rate of 3.0 m/sec, the flow rate of a treatment gas blown from each of the delivery ports for the treatment gas was determined. When the flow rate of the treatment gas was determined, nitrogen was used as the treatment gas. In addition, the flow rate of each of the two slit nozzles was determined at the positions in the axis direction, and the sum of the flow rates at the same position of the two slit nozzles was defined as the flow rate of the treatment gas at the same position in the axis direction.

(52) In addition, as a comparative example (conventional example), two of slit nozzles **7**, which had no flow-control plate, were prepared. Such two slit nozzles **1** were arranged so that positions of the delivery ports (slits) are the same in the threading direction of the steel strip in the same way as in the case of slit nozzles **3** (example of the disclosed embodiments), except that slit nozzles **7** were used instead of slit nozzles **3**. The treatment gas was fed into the slit nozzles **7** in the same way as in the case of the example of the disclosed embodiments, and the flow rate of the treatment gas blown through the delivery ports for the treatment gas was determined in the same way as in the case of the example of the disclosed embodiments.

(53) FIG. **8** is a graph illustrating the determination results of the flow rate of the treatment gas when the slit nozzles **3**, that is, the example of the disclosed embodiments, and the slit nozzle **7**, that is the comparative example, were used. FIG. **8** is a graph illustrating the relationship between the position in the axis direction and the flow rate of the treatment gas.

(54) Here, regarding the position in the axis direction (cm) measured along the horizontal axis in FIG. **8**, the point of 0 cm corresponds to the end of the delivery port on the side of the feeding port for the treatment gas of the slit nozzle into which the treatment gas was fed at a flow rate of 1.5 m/sec. In addition, the point of 70 cm corresponds to the end of the delivery port on the side of the open end of the inner tube of the relevant slit nozzle.

(55) As illustrated in FIG. **8**, in the case where the slit nozzle of the comparative example was used, a variation (maximum value–minimum value) in the flow rate of the treatment gas depending on the position in the axis direction was 0.42 m/sec. In contrast, in the case where the slit nozzle of the example of the disclosed embodiments was used, it was possible to decrease a variation (maximum value–minimum value) in the flow rate of the treatment gas depending on the position in the axis direction to 0.17 m/sec.

Example 3: Evaluating the Manufacturing of a High-Silicon Steel Strip

(56) After a silicon steel strip (having a thickness of 100 μm , a width of 600 mm, a Si concentration of 3.4 mass %, and Young's modulus of 210 GPa (room temperature) had been prepared, a high-silicon steel strip having a silicon content of 6.5 mass % was manufactured by using the continuous manufacturing line illustrated in FIG. **5**.

(57) In the manufacturing of the high-silicon steel strip of the example of the disclosed embodiments, two of slit nozzles **3** (example of the disclosed embodiments) in Example 1 were arranged for the steel strip on each of the front-surface side and back-surface side of the steel strip such that the slit nozzles adjacent to each other were arranged so that their feeding ports for a treatment gas faced in different (opposite) directions from each other in the siliconizing treating furnace of the continuous manufacturing line in FIG. **5**. In addition, as in the case of Example 2, the treatment gas (treatment gas containing silicon tetrachloride) was fed through the feeding port for the treatment gas into one of the two slit nozzles adjacent to each other in the threading direction at a flow rate of 1.5 m/sec and into the other at a flow rate of 3.0 m/sec.

(58) In addition, the high-silicon steel strip of the comparative example was manufactured in the same way as in the case of the example of the disclosed embodiments, except that the slit nozzles **7** were used instead of the slit nozzles **3**.

(59) FIG. **9** illustrates the determination results of the deviation of the Si concentration (mass %) in

the surface layer of the manufactured high-silicon steel strip depending on the position in the width direction. Here, the deviation of the Si concentration (mass %) in the surface layer was defined as the difference from the Si concentration at a reference position, where the central position in the width direction was defined as the reference position (having a deviation of 0 mass %). The Si concentration in the surface layer was determined by performing X-ray fluorescence spectrometry. (60) From the results illustrated in FIG. 9, it was clarified that, in the case where the slit nozzles of the comparative example were used, the deviation (maximum value–minimum value) of the Si concentration (mass %) in the surface layer was 0.55 mass %. In contrast, in the case where the slit nozzles of the example of the disclosed embodiments were used, it was possible to decrease the deviation (maximum value–minimum value) of the Si concentration (mass %) in the surface layer to 0.10 mass %.

(61) From the results of the examples described above, it was clarified that, according to the disclosed embodiments, it is possible to provide a slit nozzle having a double-tube structure with which it is possible to decrease a variation in the flow rate of a blown gas depending on the position in the axis direction. In addition, it was clarified that, by using the slit nozzle according to the disclosed embodiments, it is possible to stably manufacture a high-silicon steel strip having a small variation in the Si concentration depending on the position in the width direction.

Claims

1. A slit nozzle having a double-tube structure, the slit nozzle comprising: an outer tube having a delivery port for a treatment gas in an axial direction and a closed end; an inner tube having a feeding port for the treatment gas on one end and an open end that is another end inside the closed end of the outer tube, the treatment gas configured to be fed through the feeding port and blown through the delivery port; and a flow-control plate disposed between the open end of the inner tube and an end of the delivery port, the flow-control plate closing a gap between the inner tube and the outer tube, wherein an opening is formed in a plane in which the flow-control plate is disposed only in a range of the flow-control plate of 27.5° or more and 332.5° or less in terms of a central angle with respect to a reference line passing through an axis of the outer tube and a central position in a width direction of the delivery port, wherein an area fraction, which is an area of the opening divided by an area of the range of the flow-control plate of 27.5° or more and 332.5° or less in terms of the central angle with respect to the reference line, is 55% or more and 75% or less, and wherein a width of the delivery port, in a direction perpendicular to the axis of the outer tube, is 5 mm or more and 20 mm or less and is 15% or less of an outer diameter of the outer tube.
2. The slit nozzle according to claim 1, wherein the flow-control plate is disposed at the open end of the inner tube.
3. The slit nozzle according to claim 2, wherein the opening is formed symmetrically with respect to the reference line in the plane in which the flow-control plate is disposed.
4. A method for manufacturing a high-silicon steel strip including a siliconizing treating method using a plurality of slit nozzles according to claim 3, the method comprising: arranging the slit nozzles in a threading direction of a steel strip in a siliconizing treating furnace such that adjacent slit nozzles or adjacent groups of slit nozzles adjacent to each other in the threading direction are arranged so that feeding ports for the treatment gas of the slit nozzles face opposite directions from each other; and feeding the treatment gas through the feeding ports for the treatment gas into the slit nozzles and blowing the treatment gas through the delivery ports for the treatment gas of the slit nozzles onto the steel strip during transport, wherein the treatment gas contains silicon tetrachloride.
5. A method for manufacturing a high-silicon steel strip including a siliconizing treating method using a plurality of slit nozzles according to claim 2, the method comprising: arranging the slit nozzles in a threading direction of a steel strip in a siliconizing treating furnace such that adjacent

slit nozzles or adjacent groups of slit nozzles adjacent to each other in the threading direction are arranged so that feeding ports for the treatment gas of the slit nozzles face opposite directions from each other; and feeding the treatment gas through the feeding ports for the treatment gas into the slit nozzles and blowing the treatment gas through the delivery ports for the treatment gas of the slit nozzles onto the steel strip during transport, wherein the treatment gas contains silicon tetrachloride.

6. The slit nozzle according to claim 1, wherein the opening is formed symmetrically with respect to the reference line in the plane in which the flow-control plate is disposed.

7. A method for manufacturing a high-silicon steel strip including a siliconizing treating method using a plurality of slit nozzles according to claim 6, the method comprising: arranging the slit nozzles in a threading direction of a steel strip in a siliconizing treating furnace such that adjacent slit nozzles or adjacent groups of slit nozzles adjacent to each other in the threading direction are arranged so that feeding ports for the treatment gas of the slit nozzles face opposite directions from each other; and feeding the treatment gas through the feeding ports for the treatment gas into the slit nozzles and blowing the treatment gas through the delivery ports for the treatment gas of the slit nozzles onto the steel strip during transport, wherein the treatment gas contains silicon tetrachloride.

8. A method for manufacturing a high-silicon steel strip including a siliconizing treating method using a plurality of slit nozzles according to claim 1, the method comprising: arranging the slit nozzles in a threading direction of a steel strip in a siliconizing treating furnace such that adjacent slit nozzles or adjacent groups of slit nozzles adjacent to each other in the threading direction are arranged so that feeding ports for the treatment gas of the slit nozzles face opposite directions from each other; and feeding the treatment gas through the feeding ports for the treatment gas into the slit nozzles and blowing the treatment gas through the delivery ports for the treatment gas of the slit nozzles onto the steel strip during transport, wherein the treatment gas contains silicon tetrachloride.
